YUKAI WANG

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EDUCATION

Shanghai Jiao Tong University Sept. 2023 – June 2027 (expected)
B.Eng. in Computer Science and Technology; B.S. in Bioinformatics (dual degree)
Average: **90.15/100** | Rank: **9/116**
National Scholarship (top 2%) 2025, 2026
Zhiyuan Honors Scholarship (top 10%) 2023, 2024

PUBLICATIONS & PREPRINTS

- SMMBench: A Benchmark for Source-Distributed Multimodal Agent Memory
  Huacan Chai, **Yukai Wang**, Yingxuan Yang, Dan Peng, Yuanyi Song, Zhihui Fu, Weiwen Liu, Jianghao Lin, Jun Wang, Weinan Zhang.
  Advances in Neural Information Processing Systems (NeurIPS), 2026. Accepted. [arXiv] [Code] [Data]
- Retrieval-Driven Memory Reconsolidation for Long-Term LLM Agents
  Yuanyi Song, **Yukai Wang**, Xinbei Ma, Zhihui Fu, Jianghao Lin, Weiwen Liu, Jun Wang, Huarong Deng, Yong Yu, Weinan Zhang.
  International Conference on Learning Representations (ICLR), 2027. Under review. [arXiv]

RESEARCH EXPERIENCE

University of Notre Dame | Undergraduate Researcher (on-site) July 2026 – Present
EviSkill-Rerank — Evidence-Conditioned Skill Retrieval First author | **Manuscript in preparation**
Advisor: Meng Jiang, Associate Professor, Department of Computer Science and Engineering

- Independently designed, implemented, and evaluated EviSkill-Rerank to distinguish skills with similar capabilities but different suitability for a user's request.
- Combined FactCG assessments of supporting and conflicting evidence from complete skill blocks with Qwen3 relevance scores and diversity-aware reranking.
- On 1,235 SameCapRisk-Bench queries, exploratory evaluation improved Recall@1 by **3.0–4.0 pp** over reproduced SkillContrast across SkillSight/SkillRet and 0.6B/4B rerankers with shared fusion weights.

APEX Lab, Shanghai Jiao Tong University | Undergraduate Researcher (on-site) Jan. 2026 – May 2026
SMMBench — Source-Distributed Multimodal Agent Memory Second author
Advisor: Weiwen Liu, Associate Professor, School of Computer Science

- Constructed the dataset: **1,877** multimodal samples drawn from **264** independent sources.
- Implemented five task pipelines: single-/multi-hop QA, conflicts, preferences, and function calling.
- Conducted the main experiments across 13 memory/retrieval methods and contributed to manuscript drafting; the best baseline's aggregate score was 0.493 versus 0.747 with gold evidence.

APEX Lab, Shanghai Jiao Tong University | Undergraduate Researcher (on-site) Apr. 2026 – Sept. 2026
REALM — Self-Evolving Agent Memory Second author
Advisor: Weiwen Liu, Associate Professor, School of Computer Science

- Co-designed and implemented REALM, integrating heterogeneous memory graphs, adaptive search, and retrieval-driven memory reconsolidation.
- Conducted benchmark experiments; accuracy exceeded the strongest evaluated baselines: **75.97%** on LoCoMo (+7.17 pp) and **65.11%** on LongMemEval-S (+1.31 pp).
- Contributed to manuscript drafting; ablations isolated 2.01 and 2.13 pp gains from reconsolidation.

TECHNICAL SKILLS

AI/ML Tools: PyTorch, Hugging Face Transformers, LangChain, ChromaDB, scikit-learn.
Programming & Tools: Python, C/C++, Java, JavaScript, SQL, Bash; Linux, Git, Jupyter, LaTeX.
Web & Data Tools: Vue.js, HTML/CSS, Django, Flask, REST APIs; NumPy, SciPy, Matplotlib.